

ASSESSMENT TOOL 1

Concept Mapping

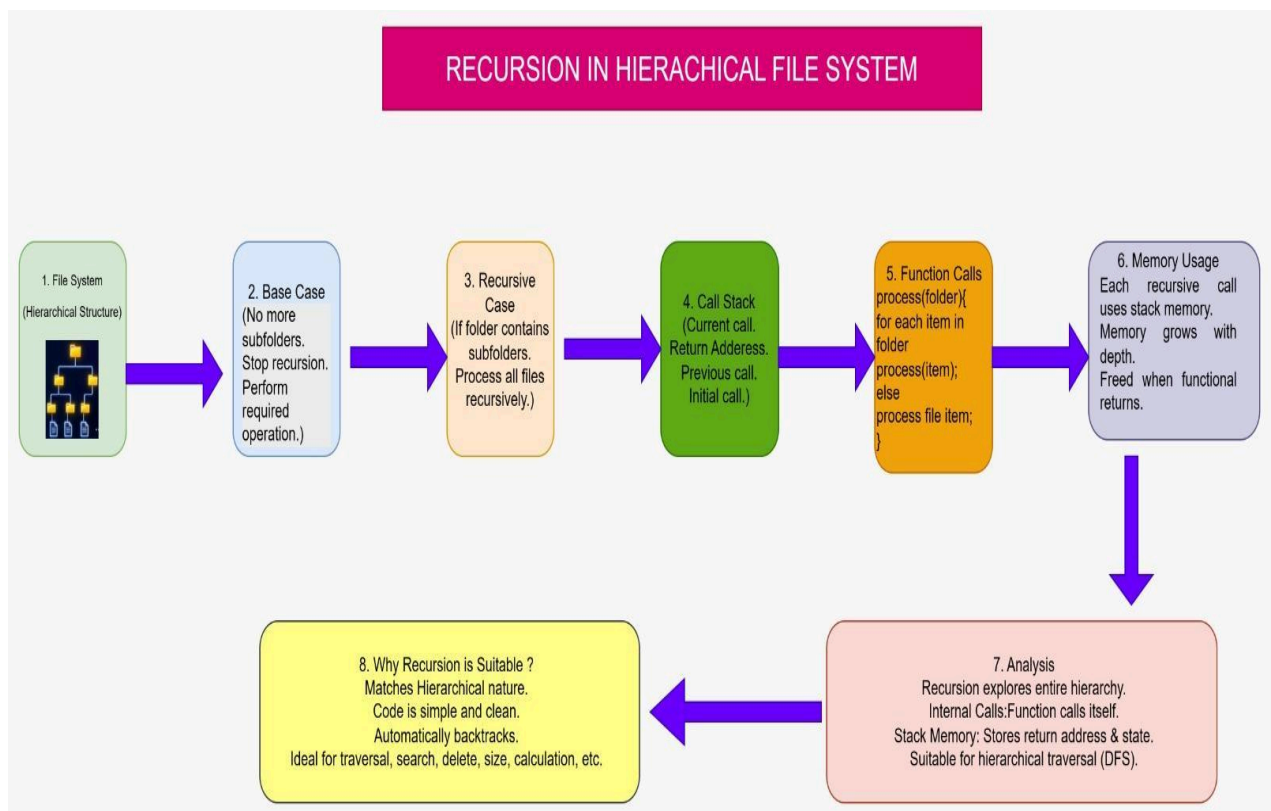
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Course Code: CSA0312

Course Name: Data Structures

7) A file system contains nested folders and files organized hierarchically. Construct a concept map linking Recursion → Base Case → Recursive Case → Call Stack → Function Calls → Memory Usage. Analyse how recursion operates internally and Explain the relationship between recursive calls and stack memory. Justify why recursion is suitable for solving



hierarchical problems.

8) A program processes complex nested data structures and requires efficient computation. Construct a concept map connecting Recursion → Call Stack → Memory Usage → Iteration → Loop Control → Efficiency. Compare the two approaches in terms of execution flow, memory consumption, and performance. Analyse their advantages and limitations, and justify which approach is more efficient in different scenarios.

